

Series	Display	Memory	µCode ROM	Expansion	Announced
700	Nixie Tubes (old)	Core (old)	Wire Weave (old)	n/a	Feb 1969
500	Nixie Tubes (old)	256x1 MOS (newer)	Wire Weave (old)	n/a	Jan 1971
520	Nixie Tubes (old)	1Kx1 MOS (newest)	Wire Weave (old)	n/a	Mid 1971?
600 early	9-Segment (new)	1Kx1 MOS (newest)	Wire Weave (old)	MOS ROM	Nov 1971
600 late	9-Segment (new)	1Kx1 MOS (newest)	MOS ROM (new)	MOS ROM	
400	9-Segment (new)	256x1 MOS*	MOS ROM (new)	EPROM (new)†	Late 1972?

* Lower capacity RAM may have been used simply because higher capacity was not needed (cost savings).

† MOS ROM expansion boards were also made.

Editorial

The 500/600 Series were a major departure from the 700, not only in technology. The 500/600 was a significant redesign in terms of user-friendliness, providing a human interface that was much more intuitive. In addition, the calculator portion was easier to use - before even resorting to programming. Some examples of these improvements are:

- 16 reserved registers, cannot be overwritten by program steps.
- The addition of one-key (one step) access/update of those 16 registers.
- Easier setup of the 16 register keys (no more abstract binary toggle switches, instead having radio buttons with operator mnemonics).
- Built-in printer that can list programs with mnemonics, as well as print calculation results, and trace programs or calculator keyboard operations.

The 500/600 line did use a completely different set of program codes from the 700, and so software could not be directly moved over. However, the 500 and 600 were software upward-compatible within the family, and would be backward compatible if a small number of advanced program codes (INDIR, I/O codes) were avoided.

The 500 supported only one peripheral, the mark-sense card reader. The interface was electrically incompatible with the 600 (and 700). The 600 supported the full range of peripherals that the 700 supported - and while the interfaces were electrically compatible, in most cases different peripheral models were required (typically to account for the difference in program codes between 600 and 700).

Wang's strategy seemed to revolve around the 700 being some sort of super-calculator, and the 500/600 line being the "economy" models. That is not an obvious ranking, though. The 700 did have two displays, the RECALL RESIDUAL function to double the precision (if one did the extra programming), and a few more program steps (which is easily offset by the powerful 500/600 one-step register functions). But, the 500/600 had built-in trig functions plus the other features mentioned (including costing less). In hindsight, it might appear that the 700 – with its old and expensive (low profit margin) technology – was more of a boat-anchor to Wang than a savior. It would be interesting to see how many applications used on the 700 could not have been ported to a 500/600.

NOTE: The editor is a software/hardware engineer that learned to program calculators on 520/600 models, and only later was exposed to the 700 as a comparison. While this influences opinions, it should not affect the comparisons of features.